VWO Login Dashboard — Test Plan

Master Test Plan · app.vwo.com · Version 1.0

Status: **DRAFT — pending human review.** Not approved until a QA owner signs off.

1. Test Plan Identifier & Metadata

Plan Name	VWO Login Dashboard — Master Test Plan
Version	1.0
Author	QA Engineering Team
Date	2026-11-02
Target Release / Version	VWO Login Dashboard v1.0 (app.vwo.com)
Document Type	IEEE 829 / Agile QA Test Plan

2. Introduction & Objectives

System Under Test (SUT)

The VWO Login Dashboard at **app.vwo.com** is the entry point to the Visual Website Optimizer platform. It provides email/password authentication, session management, optional multi-factor authentication (MFA), enterprise Single Sign-On (SSO), password recovery, and a branded, accessible login experience with Light/Dark Mode.

Goals of the Testing Effort

- Verify the login dashboard meets all functional, security, accessibility, and performance requirements in the PRD.
- Gate the release by confirming no blocker or critical defects remain.
- Validate compliance with GDPR/CCPA, OWASP authentication guidelines, and WCAG 2.1 AA.
- Confirm the login flow transitions correctly into the VWO core platform dashboard.
- Mitigate security, performance, and usability risks before production sign-off.

3. Scope of Testing

In-Scope

Area	Coverage
Authentication	Email/password login, validation, error handling, session management
Security	Encryption, HTTPS enforcement, rate limiting, secure token/session handling, password storage
Password Management	Forgot password flow, email-based recovery, password complexity requirements
MFA / SSO	Optional 2FA, SAML/OAuth enterprise SSO, social login (Google, Microsoft)
UX / UI	Responsive design, auto-focus, clickable labels, loading states, Light/Dark Mode, branding
Accessibility	ARIA labels, keyboard navigation, screen reader support, high contrast mode, WCAG 2.1 AA
Performance	Page load < 2s, concurrent login load, scalability, 99.9% availability, CDN
Integrations	Dashboard transition, analytics tracking, customer support systems, marketing tools
Compliance	GDPR, CCPA, enterprise security policies and audit trails
Platforms	Desktop browsers (Chrome, Firefox, Safari, Edge), mobile web (iOS/Android)

Out-of-Scope

Item	Justification
Biometric authentication (fingerprint/facial)	Listed under Future Enhancements, not part of v1.0
Adaptive/risk-based authentication	Future enhancement
Progressive Web App (PWA)	Future enhancement
A/B testing of the login experience	Optimization phase, not core v1.0 functionality
Post-login core platform features	Beyond the login dashboard boundary; covered by separate platform test plans

4. Types of Testing

Type	Description	Required
Functional Testing	Verify login, validation, error handling, password reset, and navigation flows	Yes
Integration Testing	SSO, social login, analytics, support, and dashboard transition integrations	Yes
Regression Testing	Ensure existing features (Remember Me, registration link, announcements) remain intact	Yes
API Testing	Authentication endpoints, token generation, session validation, rate limiting	Yes
Security Testing	Brute-force protection, encryption, session hijacking, OWASP checks, penetration testing	Yes
Performance / Load	Sub-2s load, thousands of concurrent logins, 99.9% uptime	Yes
Usability Testing	Login friction, error recovery, onboarding, user satisfaction	Yes
Accessibility Testing	WCAG 2.1 AA, keyboard/screen reader, high contrast	Yes
Compatibility Testing	Cross-browser and cross-device rendering and behavior	Yes
Compliance Testing	GDPR/CCPA handling, enterprise audit requirements	Yes

5. Approach / Methodology

Testing Techniques

- **Black-box Testing** for functional and UI flows (equivalence partitioning on email/password inputs).
- **Boundary Value Analysis** for password length, complexity thresholds, and rate-limit thresholds.
- **Equivalence Partitioning** for valid/invalid email formats and credential states.
- **White-box Testing** for session/token logic and server-side validation where code access is available.
- **Exploratory Testing** for edge cases in error recovery and responsive breakpoints.
- **Negative Testing** for invalid credentials, malformed inputs, expired tokens, and blocked accounts.

Automation vs. Manual Split

Area	Approach
Regression suite (login, validation, password reset)	Automated (Playwright/Selenium)
API authentication / rate limiting	Automated (Postman/Jest + load tools)
Accessibility audits	Automated (axe) + manual (screen reader)
Usability, visual design, error recovery	Manual
Security / penetration testing	Manual + automated scanners

6. Test Environment & Tools

Environments

Environment	Purpose
Staging	Primary functional, integration, and regression testing
Pre-production / UAT	Final validation, performance soak, UAT
Production (post-release smoke)	Deployment smoke verification

Platforms

Category	Coverage
Browsers	Chrome (latest), Firefox (latest), Safari (latest), Edge (latest)
Mobile	iOS Safari, Android Chrome — responsive/touch validation
OS	Windows 10/11, macOS, Linux (as applicable)

Tools

Purpose	Tool
UI Test Automation	Playwright, Selenium WebDriver
API Testing	Postman, REST Assured
Performance/Load	JMeter, k6, Gatling
Accessibility	axe-core, WAVE, NVDA/VoiceOver
Security	OWASP ZAP, Burp Suite
Test/Defect Management	Jira, TestRail, Zephyr

7. Test Deliverables

Deliverable	Description
Test Cases	Detailed functional, integration, security, and accessibility cases
Test Data	Valid/invalid credentials, SSO identities, password-reset tokens, rate-limit datasets
Traceability Matrix (RTM)	Requirement-to-test-case mapping covering all PRD items
Test Execution Reports	Per-cycle execution status, pass/fail counts, coverage metrics
Defect/Bug Reports	Logged defects with severity, priority, steps to reproduce
Automation Scripts	Regression and API test scripts
Sign-off Report	Final quality summary and release recommendation

8. Schedule & Milestones

Phase	Activity	Exit Condition
1. Test Planning	Finalize plan, scope, RTM	Plan approved
2. Test Case Design	Author cases, prepare data	Cases reviewed & baselined
3. Environment Setup	Configure staging, tools, accounts	Environment ready
4. Execution Cycle 1	Functional + integration	Critical/blocker defects triaged
5. Execution Cycle 2	Regression + security + accessibility	No open critical/high
6. Performance/Load	Load, soak, scalability tests	Performance targets met
7. UAT	Stakeholder validation	UAT sign-off
8. Bug Fixing & Retest	Defect remediation	Defect backlog closed
9. Sign-off	Final report and release decision	Exit criteria met

9. Roles & Responsibilities

Role	Responsibility
QA Lead	Plan ownership, coordination, sign-off
QA Engineers	Test case authoring, execution, defect reporting
Automation Engineers	Build/maintain automation and performance suites
Security Engineers	Security/penetration testing, compliance validation
Developers	Defect fixes, unit tests, build readiness
Product Owner / PM	Requirement clarification, prioritization, UAT
DevOps / IT	Environment provisioning, deployments, monitoring
Stakeholders	UAT participation, release acceptance

10. Risk Management & Mitigation

Risk	Impact	Likelihood	Mitigation
Environment instability	Delays execution	Medium	Dedicated staging; smoke check before cycles
Scope creep / late requirements	Test gaps	Medium	Scope freeze; change control
Delayed builds	Compressed testing	Medium	Incremental builds; phased testing
SSO/third-party unavailability	Blocked integration tests	Medium	Stubbed/mocked IdPs; vendor sandboxes
Security vulnerabilities	Release blocker	High	Early security review; automated scanning
Performance under load	Failed SLAs	Medium	Early load tests; auto-scaling validation
Test data privacy exposure	Compliance breach	Medium	Synthetic data; masked PII
Automation flakiness	Reduced confidence	Medium	Stable selectors; retry and triage policy

11. Pass/Fail Criteria & Exit Criteria

Entry Criteria

- Test plan approved and RTM baseline available.
- Staging environment provisioned and stable.
- Build deployed with release notes and known-issues list.
- Test data and accounts (including SSO/social logins) ready.
- Core login flow demonstrably working (smoke test passed).

Exit Criteria

- **Login Success Rate:** ≥ 95% successful authentication attempts.
- **Page Load Time:** Login page loads in under 2 seconds on standard connections.
- **Availability:** 99.9% uptime validated under load.
- **Defects:** Zero open Blocker or Critical defects; all High-severity defects fixed or waived with documented justification.
- **Test Coverage:** 100% of in-scope requirements mapped in the RTM and executed.
- **Regression:** Full regression suite passing.
- **Security:** No successful brute force/unauthorized access; 100% compliance with security audit requirements.
- **Accessibility:** WCAG 2.1 AA compliance confirmed.
- **Compliance:** GDPR/CCPA and enterprise audit requirements verified.
- **Sign-off:** Formal sign-off from QA Lead, Product Owner, and Security.

Appendix A — Requirement Traceability (Summary)

PRD Requirement	Test Type	Status
Email/password login + validation	Functional, API	To be authored
Session management & timeouts	Functional, Security	To be authored
MFA (2FA)	Functional, Security	To be authored
SSO (SAML/OAuth) + social login	Integration, Security	To be authored
Forgot password / recovery	Functional, API	To be authored
Responsive design & mobile UX	Compatibility, Usability	To be authored
Accessibility (WCAG 2.1 AA)	Accessibility	To be authored
Light/Dark Mode & branding	Functional, Usability	To be authored
Encryption, HTTPS, secure storage	Security	To be authored
Rate limiting / brute force protection	Security, API	To be authored
Performance (sub-2s, concurrency, 99.9%)	Performance	To be authored
Analytics & support integrations	Integration	To be authored
GDPR/CCPA & enterprise compliance	Compliance	To be authored